

YA'AN, China

WMO: 562870

Lat: 29.9830N

Lon: 102.9966E

Elev: 2064

StdP: 13.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.5	36.5	27.6	23.2	41.3	30.1	26.1	41.4	10.2	46.0	9.1	46.3	1.8	40	0.308

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	92.0	76.8	89.8	75.9	87.6	74.7	79.1	88.7	77.8	86.6	76.5	84.5	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	149.2	85.0	75.4	143.6	83.0	74.3	138.5	81.6	44.7	89.1	43.2	86.9	41.9	84.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.9	7.6	31.9	95.4	2.1	1.9	30.3	96.8	29.1	98.0	27.9	99.1	26.3	100.5		
(5)			30.5	81.3	2.5	1.4	28.7	82.3	27.2	83.1	25.8	83.9	24.0	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	44.2	48.0	55.8	64.8	71.1	75.5	79.3	78.9	71.6	63.9	55.8	47.0	(6)
(7)		DBStd	13.13	4.51	5.56	6.63	6.62	4.98	4.52	4.11	4.69	4.94	4.72	5.28	3.92	(7)
(8)		HDD50	422	186	96	19	1	0	0	0	0	0	0	11	109	(8)
(9)		HDD65	2421	644	476	292	83	8	0	0	4	78	277	559		(9)
(10)		CDD50	5194	7	40	200	446	654	765	908	895	649	430	185	15	(10)
(11)		CDD65	1718	0	0	8	77	197	315	443	430	203	44	1	0	(11)
(12)		CDH74	11743	0	0	40	417	1058	2036	3689	3523	890	90	0	0	(12)
(13)		CDH80	3829	0	0	3	91	261	622	1326	1300	217	9	0	0	(13)
(14)	Wind	WSAvg	2.8	2.3	2.5	3.0	3.2	3.5	3.4	3.4	3.3	2.6	2.3	2.1	2.2	(14)
(15)	Precipitation	PrecAvg	67.9	0.9	1.2	2.0	3.3	5.2	7.0	14.6	17.5	8.4	4.3	2.3	0.9	(15)
(16)		PrecMax	93.2	2.5	2.8	5.4	7.4	12.2	16.1	29.8	31.9	24.3	6.9	5.7	1.5	(16)
(17)		PrecMin	47.5	0.1	0.4	0.6	1.1	1.8	3.7	5.8	4.6	3.0	1.9	0.9	0.3	(17)
(18)		PrecStd	12.2	0.5	0.6	1.0	1.5	2.2	3.1	6.0	7.0	4.6	1.2	1.1	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	67.7	78.4	86.3	88.9	91.8	94.8	95.1	90.1	80.8	71.9	60.1	(19)
(20)			MCWB	48.9	53.9	61.3	67.0	69.6	73.6	77.3	78.4	76.0	67.7	61.0	50.6	(20)
(21)		2%	DB	55.8	62.9	73.4	82.6	85.9	88.9	92.0	92.2	85.5	76.4	68.3	57.0	(21)
(22)			MCWB	46.8	52.1	59.2	65.8	68.3	72.9	77.3	77.5	73.5	66.2	59.7	49.0	(22)
(23)		5%	DB	52.8	59.4	69.4	79.0	82.9	86.4	89.7	89.9	82.2	73.1	65.2	54.5	(23)
(24)			MCWB	45.7	50.6	57.7	64.7	67.7	72.2	76.7	77.0	71.8	65.4	58.6	48.3	(24)
(25)		10%	DB	50.3	56.2	65.6	75.2	80.0	83.6	87.3	87.2	79.0	70.2	62.6	52.3	(25)
(26)			MCWB	44.7	49.5	56.2	63.5	67.0	71.6	75.9	76.0	70.6	64.1	57.5	47.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	54.9	64.5	70.3	73.1	77.4	81.0	80.9	78.6	70.5	62.6	52.8	(27)
(28)			MCD B	58.6	64.6	74.6	81.9	83.0	86.1	91.1	91.1	87.0	76.7	68.8	57.1	(28)
(29)		2%	WB	47.9	53.1	60.8	67.8	71.1	75.6	79.2	79.2	75.7	67.9	60.9	51.0	(29)
(30)			MCD B	53.4	60.6	69.5	77.9	80.7	84.0	89.1	88.7	82.4	73.6	66.4	54.4	(30)
(31)		5%	WB	46.6	51.6	58.8	65.9	69.5	74.4	77.9	78.0	73.7	66.4	59.6	49.5	(31)
(32)			MCD B	51.4	57.7	67.3	75.7	79.0	82.4	87.0	86.9	79.6	71.1	63.6	52.9	(32)
(33)		10%	WB	45.4	50.0	57.1	64.2	68.2	72.9	76.6	76.7	71.9	64.9	58.4	48.2	(33)
(34)			MCD B	49.6	55.5	64.6	73.3	76.9	80.5	84.6	85.0	77.3	69.4	61.7	51.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.4	12.0	13.6	13.2	12.4	12.4	12.5	9.9	8.7	8.1	7.9	(35)
(36)			MCDBR	13.6	16.0	18.5	19.5	18.2	17.1	16.9	16.4	15.2	13.7	12.6	12.4	(36)
(37)			MCWBR	6.9	7.7	8.1	7.7	7.5	7.1	7.4	6.9	6.4	5.7	5.8	6.5	(37)
(38)		5% WB	MCD BR	11.9	14.4	16.3	17.3	15.3	14.5	15.2	15.0	13.1	11.5	10.4	9.8	(38)
(39)			MCWBR	6.3	7.1	7.6	7.5	7.0	6.7	7.1	6.7	5.8	5.3	5.2	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.622	0.658	0.714	0.707	0.676	0.636	0.672	0.700	0.698	0.683	0.660	0.623	(40)	
(41)		taud	1.577	1.543	1.493	1.553	1.651	1.768	1.695	1.632	1.615	1.597	1.572	1.597	(41)	
(42)		Ebn at Noon	184	194	196	206	213	222	213	205	197	186	174	175	(42)	
(43)		Edh at Noon	73	82	91	89	81	71	76	81	80	76	72	68	(43)	
(44)	All-Sky Solar Radiation	RadAvg	738	887	1053	1258	1286	1195	1289	1239	933	772	729	676	(44)	
(45)		RadStd	89	96	100	118	137	155	161	184	90	99	51	79	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.56	N/A	N/A	+0.88	N/A	N/A	-90	-164	+295	+175

Nomenclature: See separate page